

BST Vol 8 Ch 2 — Newton's Laws + Inertia from Substrate (v0.3, Wave 3)

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Vol 8 Chapter 2 — Newton's Laws + Inertia

Headline result

Newton's three laws + inertial mass concept emerge from substrate D_{IV}^5 via K-type Casimir framework. Inertial mass scales with substrate energy unit ($C_2 = 6$ via T2435 + T2439 RIGOROUSLY CLOSED ground state).

Substrate mechanism

- $F = ma$: substrate dynamics in classical limit (Vol 5 Lyra)
- Inertial mass = substrate K-type Casimir resonance scale
- Action-reaction (3rd law): substrate momentum conservation T2474 (Lyra Friday)

Cross-volume dependencies

- Vol 1 Ch 5 (Casimir Algebra) — C_2 substrate energy unit
- T2474 (Lyra) — momentum conservation
- Vol 5 (Lyra) — classical limit

K-audit anchor

K230 (Keeper pending).

Pedagogical walkthrough

Level 1

Newton's $F = ma$ says force causes acceleration; "inertia" resists motion. BST predicts: inertia comes from substrate K-type Casimir $C_2 = 6$ energy structure.

Level 2-3

$F = ma$ Newton's 2nd law; inertial mass m scales with substrate energy unit $C_2 = 6$ (T2435 RIGOROUSLY CLOSED). 3rd law (action-reaction) inherits substrate momentum conservation T2474 via coset translation invariance.

Honest scope: inertial-mass-equivalent gravitational-mass equivalence (Einstein equivalence principle) requires Vol 4 (GR) cross-link.

Bibliography

1. I. Newton (1687): *Principia Mathematica*.
2. T2435 + T2439 + T2474.

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